

Jian Wei CHEONG

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Curriculum Vitae

Professional Qualifications

- 2019 - 2023 **PhD, Physics**, *Nanyang Technological University*, Singapore
2015 - 2019 **BSc, Physics, Honours**, *Nanyang Technological University*, Singapore
2010 - 2013 **Diploma, Electrical Engineering**, *Ngee Ann Polytechnic*, Singapore

Professional Experience

- 2024 - current **Postdoctoral Research Fellow**, *Nanyang Technological University*, Singapore
2023 **Project Officer**, *Nanyang Technological University*, Singapore
2012 **Intern**, *ST Electronics*, Singapore
Teaching Assistant, *Nanyang Technological University*, Singapore

Year	Course
2026	Physics Discovery Camp - Computational Nonlinear Lab
2025	Physics Discovery Camp - Computational Nonlinear Lab
2025	PH3101 Quantum Mechanics 2 (Restricted Repeat)
2024	Physics Discovery Camp - Computational Nonlinear Lab
2021	PH3101 Quantum Mechanics 2
2020	PH1199 Physics Lab 1B
2019	PH1198 Physics Lab 1A

Awards & Achievements

- 2019 **Short-speech Contest Best Presentation** (PAP701 Graduate seminar module), *Nanyang Technological University*, Singapore
2017/2018 **Dean's List** (top 5% of cohort), *Nanyang Technological University*, Singapore
2016/2017 **NTU President Research Scholar** (completing URECA), *Nanyang Technological University*, Singapore
2011 **Director's List** (top 5% of cohort), *Ngee Ann Polytechnic*, Singapore
2011 **Best Performance, Programmable Logic Device** (top student of cohort), *Ngee Ann Polytechnic*, Singapore
2010 **Best Performance, Digital Electronics & Practice** (top student of cohort), *Ngee Ann Polytechnic*, Singapore

Bibliography

Publications

1. Y. Wang, A. Pradana, [J. W. Cheong](#), and L. Y. Chew, **Enhancing performance and operational regimes of the quantum Otto refrigerator with heat bath algorithmic cooling**, *Physical Review E*, **113**(5), 054112 (2026).
2. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Enhancing the performance of quantum switch refrigeration with an information pool of qubits**, *Physical Review Research*, **8**(1), 013272 (2026).
3. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Non-Markovian refrigeration and heat flow in the quantum switch**, *Physical Review A*, **110**(2), 022220 (2024).
4. L. Y. Chew, A. Pradana, L. He, and [J. W. Cheong](#), **Stochastic thermodynamics of finite-tape information ratchet**, *European Physical Journal Special Topics* (2023).
5. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Effects of non-Markovianity on daemonic ergotropy in the quantum switch**, *Physical Review A*, **108**(1), 012201 (2023).
6. L. He, [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Effects of correlation in an information ratchet with finite tape**, *Physical Review E*, **107**(2), 024130 (2023).
7. [J. W. Cheong](#), A. Pradana, and L. Y. Chew, **Communication advantage of quantum compositions of channels from non-Markovianity**, *Physical Review A*, **106**(5), 052410 (2022).
8. L. He, A. Pradana, [J. W. Cheong](#), and L. Y. Chew, **Information processing second law for an information ratchet with finite tape**, *Physical Review E*, **105**(5), 054131 (2022).

Proceedings

1. L. Y. Chew, [J. W. Cheong](#), and A. Pradana, **Thermodynamic Functionality of Non-detailed Balance Finite-Tape Information Ratchet**, *International Conference on Geometric Science of Information, Springer Nature Switzerland*, pp. 173–181 (2025).

Talks

1. [J. W. Cheong](#)[†], A. Pradana, and L. Y. Chew, **Non-Markovian Quantum Refrigeration in the Quantum Switch**, *Quantum Thermodynamics Conference 2026 (QTD2026)*, Natal, Brazil, August 2026.
2. [J. W. Cheong](#)[†], A. Pradana, and L. Y. Chew, **Maxwell's Uncertain Demon: Achieving Nonclassical and Enhanced Refrigeration when Causality Blurs**, *Asia-Pacific Conference on Networks and Complex Systems 2026 (APCNCS 2026)*, Singapore, June 2026.

Posters

1. [J. W. Cheong](#), A. Pradana, and L. Y. Chew[†], **Enhancement of quantum processes from indefinite causal order through non-Markovianity**, *29th International Conference on Statistical Physics (STATPHYS29)*, Florence, Italy, July 2025.
2. [J. W. Cheong](#)[†], A. Pradana, and L. Y. Chew, **Non-Markovian refrigeration and heat flow in the quantum switch**, *Quantum Thermodynamics Conference 2025 (QTD2025)*, Singapore, July 2025.
3. L. Y. Chew[†], [J. W. Cheong](#), and A. Pradana, **Enhancement of quantum processes from indefinite causal order through non-Markovianity**, *XLV Dynamics Days Europe 2025 (DDE2025)*, Thessaloniki, Greece, June 2025.
4. L. Y. Chew[†], L. He, A. Pradana, and [J. W. Cheong](#), **Stochastic thermodynamics and correlation effects of finite-tape information ratchets**, *28th International Conference on Statistical Physics (STATPHYS28)*, Tokyo, Japan, August 2023.

[†]Presenting author.

Technical Experience

	Skill	Level	Comment
Programming	Python		9+ years experience, used in main work
	Julia		4+ years experience, used in main work
	R		graduate course, computational biology, data analysis
	C / C++		bachelor course
	MATLAB		bachelor course
	Haskell		personal quantum computing projects
	Racket		personal quantum computing projects
	Common Lisp		personal software projects
	Bash / sh		personal Linux projects
	Quarto		website, presentations, and reports
	LaTeX / Typst		presentations, reports, and published papers
	HTML / CSS		personal website
	basic knowledge extensive knowledge intermediate knowledge expert knowledge		

Miscellaneous Projects

- **Strain estimation for hazard forecastings before and after 2011 Japan Tohoku earthquake** by analysis of seismic GPS displacement data and computation of seismic strains from velocity fields.
ES7008 Geophysical Data Analysis, NTU
- **Variations in statistical complexity of genome sequences across species** by analysis of genome sequences from GenBank assembly with Baum-Welch algorithm.
CE7412 Computational and Systems Biology, NTU
- **Detecting adversarial attack of deep neural networks for image recognition from image complexity** by Fast Gradient Sign Method (FGSM), DeepFool, One Pixel Attack, and Jacobian-Based Saliency Map Attack (JSMA).
PH3502 Chaotic Dynamical Systems, NTU
- **Monte Carlo photon transport in multi-layered biological tissues** with Henyey-Greenstein scattering for diagnostic imaging and photon therapy.
PH4505 Computational Physics, NTU
- **Monte Carlo simulation of periodic-driven Brownian particles** with Arrhenius equation, demonstrating dissipation-driven adaptation.
PAP723 Numerical Methods for Physicists, NTU
- **Designing, programming, 3D printing, and building a hovering quadcopter drone** with Arduino, Autodesk EAGLE, Fusion360, and COMSOL Multiphysics.
Making and Tinkering Lite 1, NTU